Contents

PMSM Machines, Control, and Fault Diagnosis — Engineering Background

Industrial Electrical Appliances Lab · Mechatronics Engineering. This chapter provides the electrical-machines and control-engineering foundation for the project. It explains the motor itself, its drive/control system, each fault and how it arises, how such faults are detected today (manually and by existing automated methods), how the proposed scalogram-CNN approach improves on them, and how the PMSM compares with its relatives. It is meant to be read as the theory part of the main report (`report.md`).

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1. Why this matters in industry

Electric motors consume roughly **40–45 % of the world's electricity** and are the core actuator of almost every industrial appliance: pumps, fans, compressors, conveyors, CNC spindles, robots, and electric vehicles. An unplanned motor failure stops a production line; in many plants downtime costs **thousands of dollars per hour**. The maintenance philosophy has therefore shifted:

- **Reactive** (“run to failure”) → expensive, dangerous.
- **Preventive** (fixed schedule) → wasteful, still misses random faults.
- **Predictive / condition-based (CBM)** → monitor the machine’s signals and act *before* failure. This is a pillar of **Industry 4.0 / smart manufacturing**.

This project sits squarely in **predictive maintenance**: it listens to the motor’s own current and vibration and flags an incipient stator fault early enough to schedule a repair. Detecting an **inter-turn short** in its first seconds can prevent a burned-out winding and a full motor replacement.

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2. The PMSM machine

2.1 Construction

A Permanent-Magnet Synchronous Motor has two main parts (Fig. 1):

- **Stator** (stationary): a laminated iron core with slots carrying a **three-phase distributed winding** (phases A, B, C). Laminations reduce eddy -current losses; the winding insulation is the component that fails in an inter-turn fault.
- **Rotor** (rotating): carries **permanent magnets** (typically NdFeB) that provide the excitation flux — no rotor winding, no slip rings, no brushes.
- **Air gap**: the small magnetic clearance between stator and rotor where energy is transferred.

Two rotor topologies matter:

- **SPM (Surface-mounted PM):** magnets glued on the rotor surface; `L_d ≈ L_q` (no saliency); simple, used at low/medium speed.
- **IPM (Interior PM):** magnets buried inside the rotor; `L_d ≠ L_q` gives extra **reluctance torque** and a wide field-weakening range (preferred in EV traction).

2.2 Operating principle — synchronous rotation

When the three stator phases are fed with balanced sinusoidal currents 120° apart, their magneto-motive forces add up to a **rotating magnetic field** whose speed is set by the supply frequency and pole pairs, `n_s = 120·f / P` (rpm). The rotor magnets **lock onto** this rotating field and turn **in synchronism** with it — hence *synchronous* motor (Fig. 2). Unlike an induction motor there is **no slip** and **no rotor current**, which is the root of the PMSM’s high efficiency.

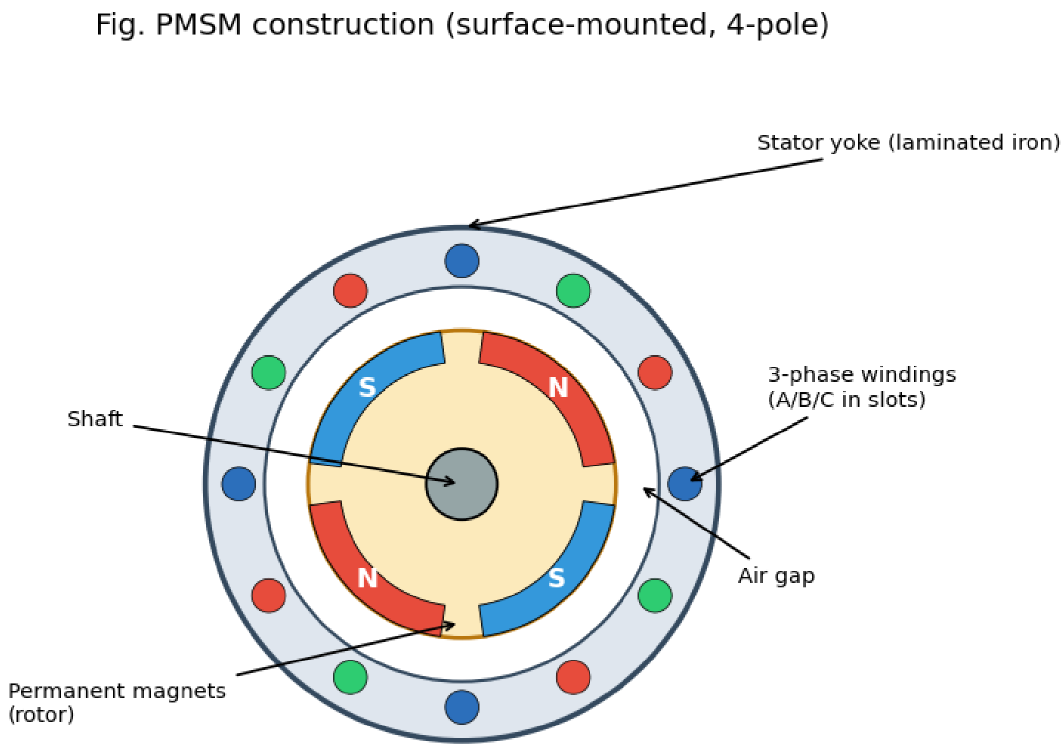

Figure 1: Fig. 1 — PMSM cross-section

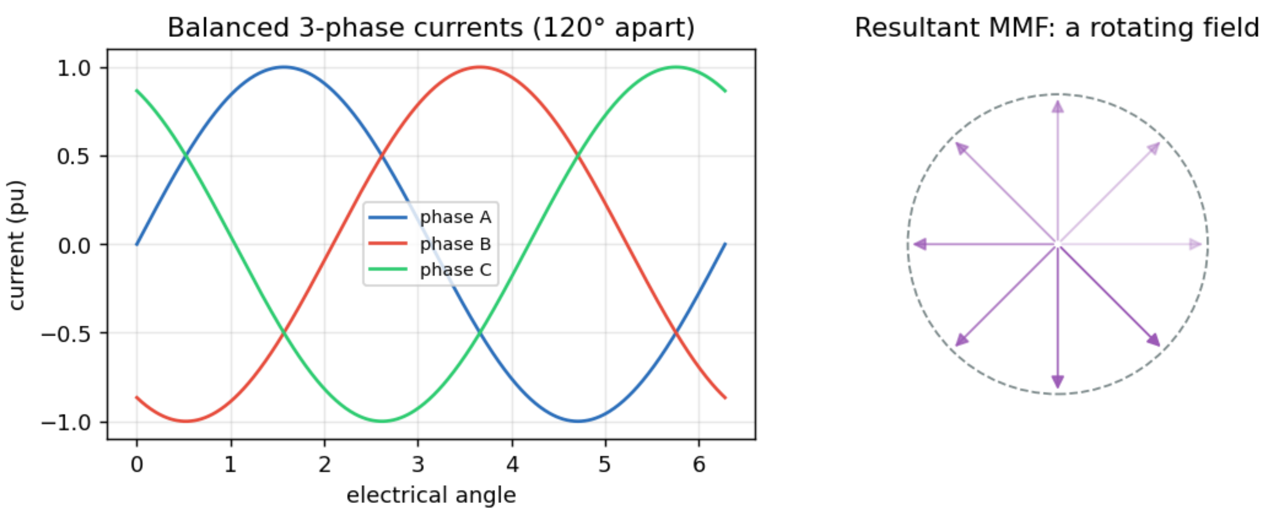

Figure 2: Fig. 2 — Rotating field from 3-phase currents

2.3 Electromagnetic torque and the d–q frame

Controlling an AC motor directly in the three-phase (abc) frame is hard because everything is sinusoidal and coupled. The **Clarke** transform collapses abc → two orthogonal stationary axes (αβ); the **Park** transform rotates αβ → a frame that spins with the rotor (d–q), where the sinusoids become **DC quantities**:

- **d-axis (direct):** aligned with the rotor flux. In a PMSM we usually force $i_d \approx 0$ (the magnets already provide the flux).
- **q-axis (quadrature):** perpendicular to the flux; torque is proportional to it, $T \approx (3/2) \cdot P \cdot \lambda_m \cdot i_q$. So **$i_q$ is the "torque knob."**

This decoupling (flux and torque made orthogonal, exactly as in a DC motor) is what makes high-performance PMSM control possible.

2.4 Back-EMF: sinusoidal (and why it matters)

A spinning rotor induces a **back-EMF** in the stator. Its shape distinguishes the PMSM (sinusoidal) from the BLDC (trapezoidal) (Fig. 3). The sinusoidal back-EMF is why PMSM drives use sinusoidal currents and **Field-Oriented Control**, and why **time–frequency (wavelet) analysis** of the current is so natural — the healthy signal is a clean fundamental, and faults perturb it in characteristic ways.

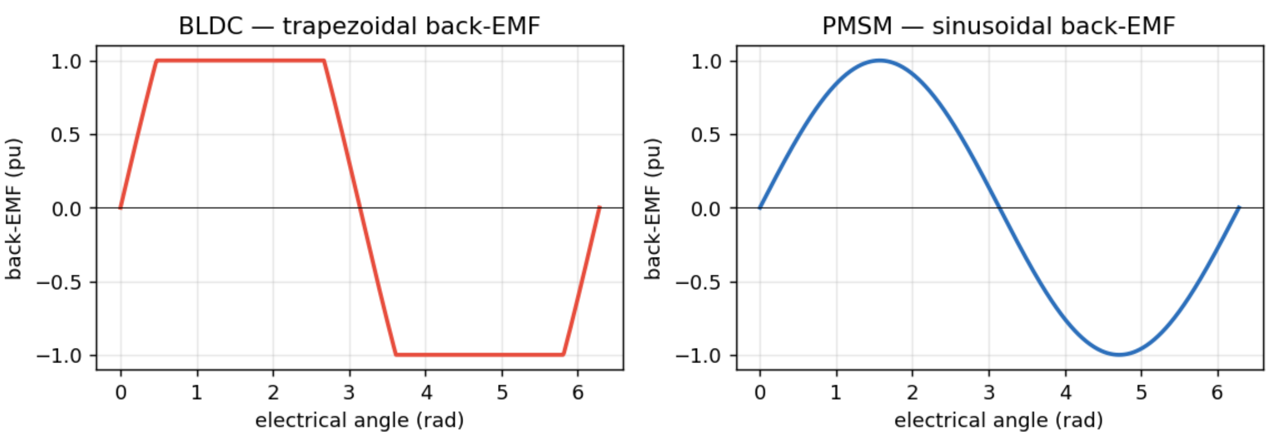

Figure 3: Fig. 3 — Back-EMF: BLDC vs PMSM

2.5 Torque–speed envelope

Below base speed the drive holds **constant torque**; above it, the back-EMF would exceed the supply, so the controller injects negative i_d to **weaken the field** and runs at **constant power** (Fig. 4). Operating point (load, speed) strongly affects the measured signals — a key reason fixed-threshold fault detectors struggle and learned features help.

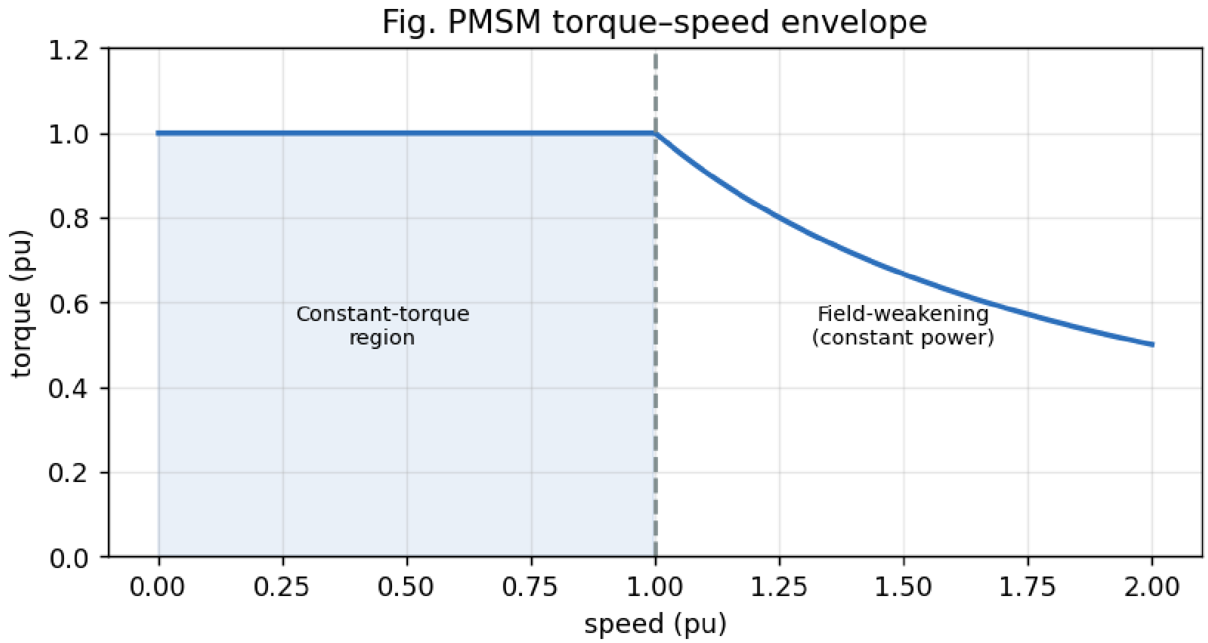

Figure 4: Fig. 4 — Torque–speed envelope

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3. PMSM vs its relatives (PMDC, BLDC, induction)

Understanding *why* the PMSM was chosen requires comparing it with neighbouring machines (Fig. 5):

Fig. PMSM vs its relatives (PMDC / BLDC / Induction)

	PMSM	PMDC	BLDC	Induction
Commutation	Electronic (inverter)	Mechanical brushes	Electronic (inverter)	Electronic/DOL
Back-EMF	Sinusoidal	—(DC)	Trapezoidal	Sinusoidal
Efficiency	Very high	Medium	High	Medium
Torque density	Very high	Low	High	Medium
Control complexity	High (FOC)	Low	Medium	Medium/High
Maintenance	Low (no brushes)	High (brushes)	Low	Low
Rotor current	None	Yes (armature)	None	Induced (slip)
Typical use	EV, robotics, servo	Toys, simple drives	Drones, fans	Pumps, industry

Figure 5: Fig. 5 — Motor family comparison

- **PMDC (brushed DC):** simplest control (voltage = speed), but mechanical **brushes/commutator** wear out, spark, and need maintenance; lower efficiency.

The PMSM is essentially “a PMDC turned inside-out” with *electronic* commutation — same easy torque control, no brushes.

- **BLDC:** also PM rotor + electronic commutation, but **trapezoidal** back-EMF, driven with six-step square currents → higher torque ripple, simpler/cheaper control. PMSM (sinusoidal + FOC) gives smoother torque and higher efficiency — better for servo/traction.
- **Induction motor:** rugged and cheap, no magnets, but relies on **rotor slip and induced rotor current** → rotor copper losses and lower efficiency/torque density. Dominant in fixed-speed industrial pumps/fans.

Summary: the PMSM combines the DC motor’s easy, decoupled torque control with brushless reliability and the highest efficiency and torque density — which is why it dominates EVs, robotics, and high-performance servo drives, and why its health monitoring is economically important.

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4. The control / drive system

A PMSM almost never runs “across the line”; it is driven by a **power-electronic inverter** under **Field-Oriented Control** (Fig. 6).

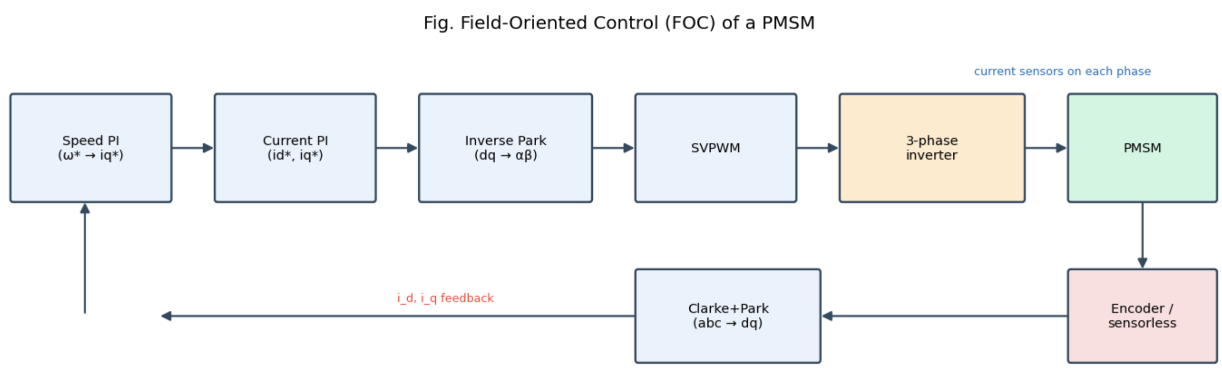

Figure 6: Fig. 6 — FOC block diagram

The loop, outer to inner:

1. **Speed controller (PI):** compares commanded vs measured speed → produces the torque command `i_q*` (with `i_d* ≈ 0`).
2. **Current controllers (PI):** regulate `i_d`, `i_q` to their references using the measured phase currents transformed into the d–q frame (Clarke + Park).
3. **Inverse Park + SVPWM:** convert the d–q voltage demands back to three-phase

and compute the switching pattern.

4. **Three-phase voltage-source inverter (Fig. 7):** six power switches (IGBT/MOSFET) chop the DC bus to synthesise the desired sinusoidal voltages.
5. **Rotor position** from an **encoder/resolver** (or a **sensorless** estimator) closes the Park transform and the speed loop.

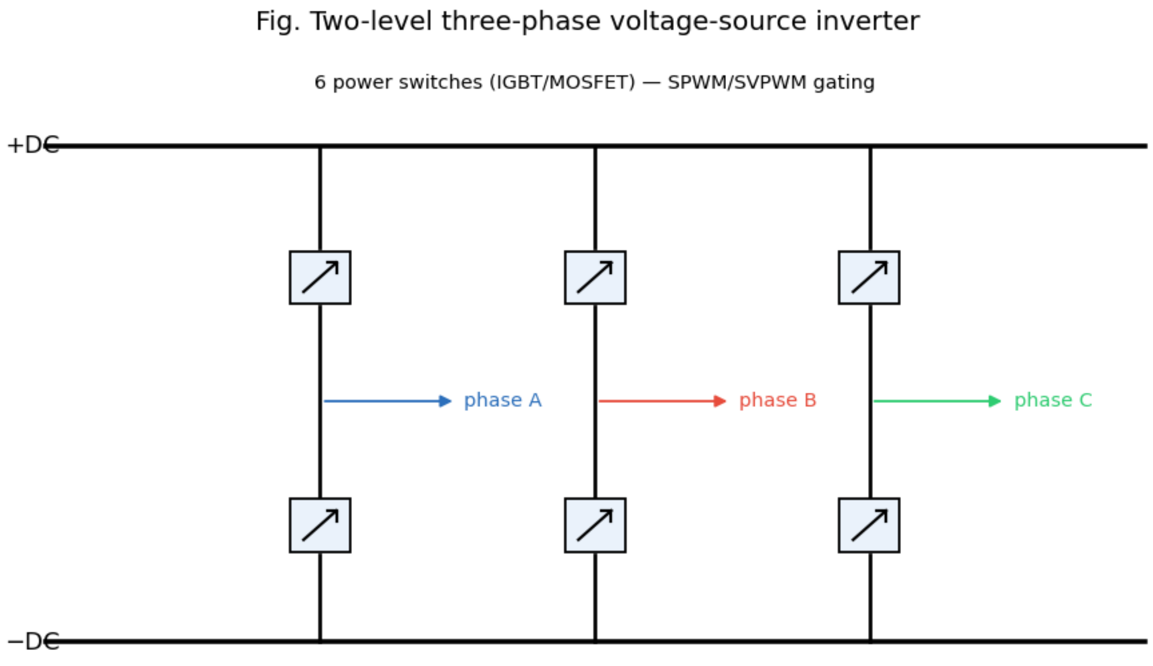

Figure 7: Fig. 7 — Three-phase inverter

Why the controller matters for diagnosis (important): the current control loop is *designed to reject disturbances*. When a fault perturbs the current, the PI controllers partially **cancel** that perturbation to keep `i_d, i_q` on target — so the fault signature in the **stator current is attenuated**. The **vibration** signal is outside this loop and responds directly to the fault. This is the physical reason our results show **vibration ≫ current** for inter-turn detection.

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5. PMSM faults — one by one

Machine faults are grouped as **electrical** (≈ 35–40 %, mostly stator winding), **mechanical** (bearings ≈ 40 %, the largest single category), and **magnetic** (demagnetization). This project targets the stator electrical fault and treats the others as context.

5.1 Inter-turn stator short circuit (ITSC) — primary fault

- **Cause:** insulation between adjacent turns of the same phase degrades from thermal ageing, voltage spikes (inverter dv/dt), moisture, vibration, or manufacturing defects.
- **Mechanism (Fig. 8):** the failed insulation creates a **shorted loop**. The rotating flux induces a large **circulating current** in those few turns (can be many times rated current), causing intense **local heating**.
- **Symptoms:** stator-current asymmetry, extra harmonics and side-bands, increased negative-sequence current, a localized hot-spot, and a characteristic vibration signature at twice the supply frequency.
- **Progression:** heat damages neighbouring insulation → the short grows turn-to -turn → phase-to-phase → ground fault → **complete winding burnout, often within minutes**. Early detection is therefore critical — exactly this project's goal.

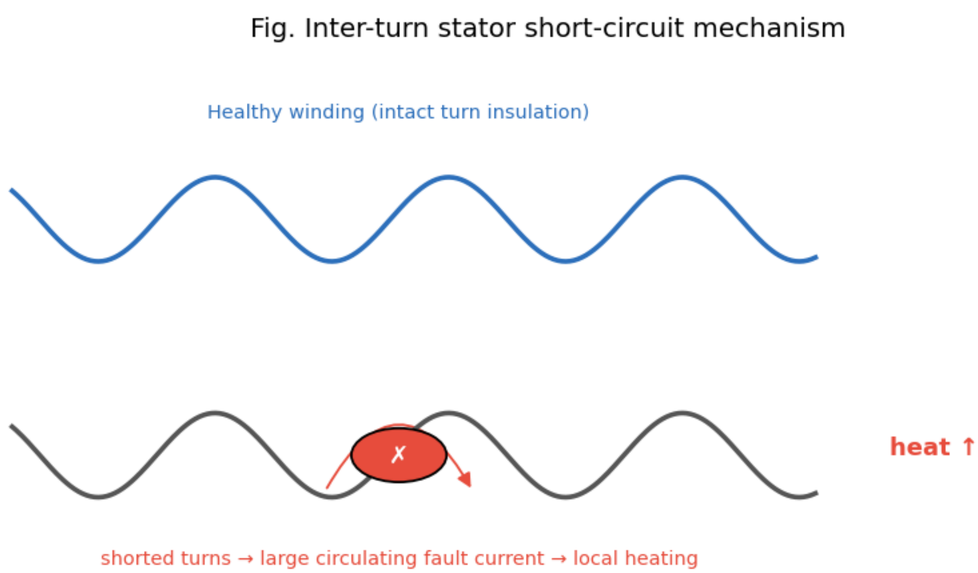

Figure 8: Fig. 8 — Inter-turn short mechanism

5.2 Demagnetization

- **Cause:** rotor magnets lose strength from over-temperature, ageing, or large demagnetising stator currents (e.g., a fault or aggressive field weakening).
- **Mechanism:** reduced PM flux λ_m lowers the back-EMF and torque constant.
- **Symptoms:** the drive draws **more current for the same torque**; **sub-harmonics** and side-bands at $f_0 \pm k \cdot f_r$ (around the rotational frequency) appear; efficiency drops.

- **Note:** absent from the public real dataset, so represented synthetically here.

5.3 Overload / thermal stress

- **Cause:** sustained operation above rated load/torque, blocked cooling, high ambient temperature.
- **Mechanism/symptoms:** raised fundamental current, extra harmonics, and rising winding temperature that **accelerates insulation ageing** — i.e., overload is often the *precursor* to an inter-turn fault. A degraded operating *state* rather than a hard fault.

5.4 Mechanical faults (context)

- **Bearings** (the most common mechanical fault): wear/pitting produce vibration at characteristic defect frequencies and, via air-gap modulation, current side-bands.
- **Eccentricity / misalignment:** uneven air gap → side-bands around the fundamental. Addressed naturally by the same scalogram+CNN method on vibration, and listed as future work.

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6. How these faults are detected today

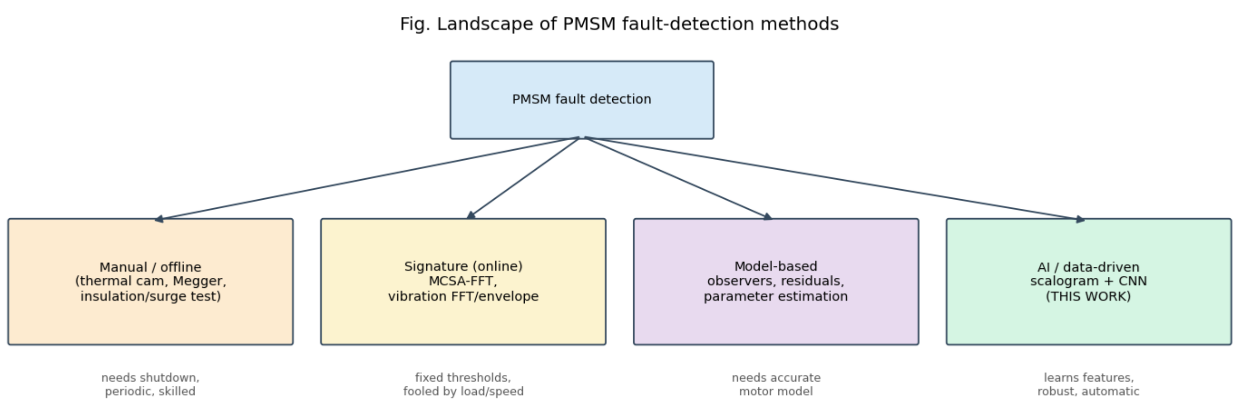

Figure 9: Fig. 9 — Fault-detection landscape

6.1 Manual / offline methods (machine stopped)

- **Visual + thermal imaging:** an infrared camera finds hot-spots — but only once heating is significant, and it needs access and a technician.
- **Insulation-resistance test (Megger) / polarization index:** DC megohmmeter checks winding-to-ground insulation. Catches gross degradation, **not** an incipient turn-to-turn short.
- **Surge / hi-pot test:** applies steep voltage to reveal turn insulation weakness — sensitive, but **offline** and potentially stressful to the winding.
- **Limitations:** require **shutdown**, are **periodic** (miss faults between inspections), and need skilled labour.

6.2 Online signature-based methods (machine running)

- **Motor Current Signature Analysis (MCSA):** take the FFT of the steady-state stator current and watch specific fault frequencies / side-bands (Fig. 10). The classic, sensor-free industrial method.
- **Vibration analysis:** accelerometer + FFT / **envelope analysis** for bearing and mechanical faults (ISO 10816 severity bands).
- **Park's Vector / instantaneous power / negative-sequence** approaches.
- **Limitations:** assume (near-)**stationary** operation; use **fixed thresholds** that are easily fooled by changing **load and speed**; the analyst must know *which* frequency to inspect; transients get **smeared** by the FFT.

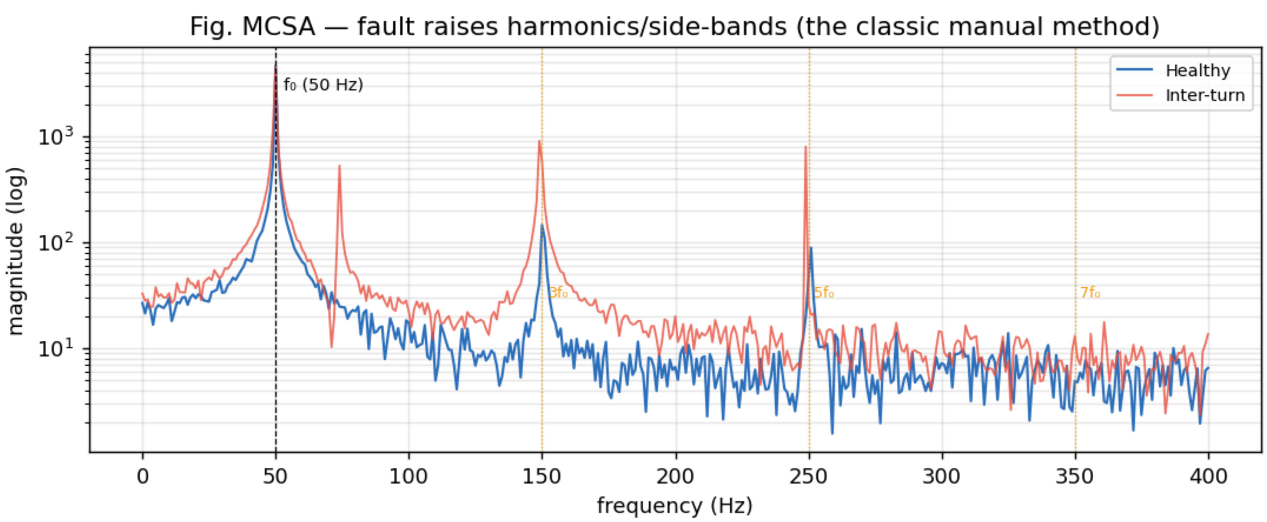

Figure 10: Fig. 10 — MCSA spectrum, healthy vs faulty

6.3 Model-based methods

- **State observers, parameter estimation, residual generation:** compare the real machine with a mathematical model; a growing residual flags a fault.
- **Limitations:** need an **accurate motor model** and parameters, which drift with temperature and ageing.

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7. How the proposed method (CWT scalogram + CNN) improves things

The new approach replaces "pick a frequency and a threshold" with "**let the network learn the fault's time–frequency fingerprint**":

1. **Captures non-stationary / transient signatures.** The Continuous Wavelet Transform keeps **time *and* frequency**, so a brief fault transient that the FFT would smear stays localized and visible (this is the core of §3–§4 of the main report).
2. **Automatic feature learning.** The CNN discovers the discriminative patterns in the scalogram itself — no expert must specify side-band frequencies.
3. **Robust to operating point.** Learned 2-D textures generalise across load/speed better than a single fixed amplitude threshold.
4. **Earlier, more sensitive detection.** Because subtle patterns are learned, the fault is flagged at lower severity than a fixed-threshold FFT alarm (Fig. 11) — buying time before burnout.
5. **Online, automatic, and sensor-reuse.** Current sensors already exist in every FOC drive; the method runs continuously with no shutdown.

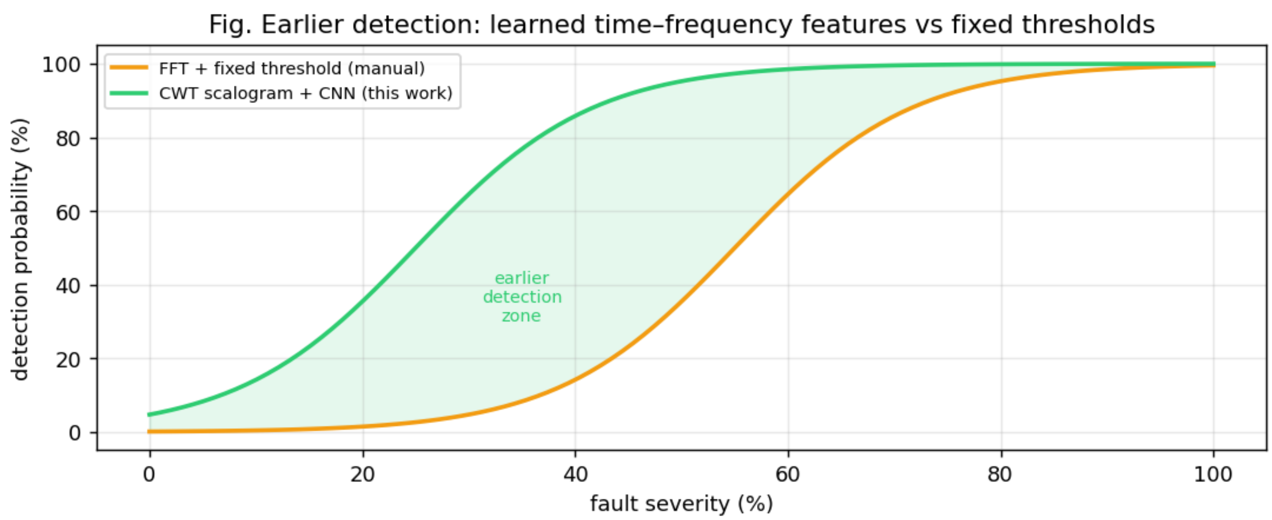

Figure 11: Fig. 11 — Earlier detection vs fixed-threshold FFT

Honest positioning. It does not replace offline safety tests (Megger/surge remain part of commissioning); it complements them as a **continuous, intelligent online monitor**. Its current limitation is data diversity (few healthy recordings), discussed in the results.

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8. Standards, safety, and economics

- **Condition-monitoring standards:** ISO 20958 (electrical signal CM of machines), ISO 10816 / 20816 (vibration severity), IEEE 1415 (motor maintenance), IEC 60034 (rotating machines).
- **Insulation classes** (IEC 60085: B/F/H) set the thermal limits whose violation drives inter-turn failure — linking overload (§5.3) to ITSC (§5.1).
- **Safety:** an undetected stator short is a fire/arc-flash hazard; early detection is a safety function, not just an economic one.
- **Economic case:** a low-cost software monitor reusing existing current sensors, preventing one burnout, typically pays for itself many times over in avoided downtime and motor-replacement cost.

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9. Link to the rest of the report

This background motivates the pipeline detailed in the main report: **signal → CWT scalogram (§3–§4) → CNN (§5) → diagnosis (§7–§8)**, with the engineering insight that **vibration carries the strongest inter-turn signature** (because the current loop suppresses it, §4) — exactly what the experimental results confirm.

All figures in this chapter are generated reproducibly by `python/figures_engineering.py` into `docs/figures/`.